

AI-DRIVEN RESUME RANKING & CANDIDATE SHORTLISTING

*Ranking candidates the way a great recruiter would —
not by keywords, but by understanding who really fits.*

HACKATHON SUBMISSION

SEMANTIC · STRUCTURAL · BEHAVIORAL

Keyword Filters Are Failing Recruiters

75%

of qualified candidates

are rejected by ATS before a human ever sees them

60%

of recruiters say

keyword systems miss the best cultural &
behavioral fits

1000+

profiles per role

a recruiter must sift through in a competitive
market

The root cause: systems match words, not meaning — and miss the forest for the trees.

Three Layers of Understanding



Semantic Similarity

Sentence-transformer embeddings compare MEANING, not words. A resume saying 'real-time streaming pipelines' matches a JD asking for 'streaming data systems' — even with zero shared words.



Structured Fit Scoring

JD parsed into structured spec: required skills, experience range, seniority level. Each candidate scored on skill coverage, experience fit, and role match — explainably.



Behavioral & Platform Signal

Leadership language, ownership phrases, impact statements, GitHub/LinkedIn activity — the signals that separate doers from job description fillers.

End-to-End Pipeline

01 JD Parser

Rule-based + optional LLM
role_title · required_skills
seniority · min_exp_years

02 Data Loader

Schema-tolerant CSV/XLSX
Column aliasing via COLUMN_MAP
Builds full_text blob per candidate

03 Embedder

all-mpnet-base-v2
Cosine similarity [0,1]
TF-IDF fallback if offline

04 Feature Extractor

skill_match · exp_match
behavioral_signal
platform_activity_score

05 Hybrid Scorer

Weighted combination → 0-100
Plain-English explanation
per candidate

06 LLM Re-ranker

Claude as senior recruiter
Top-N only (cost-efficient)
Reasoning string per candidate



Hybrid Score Breakdown

Semantic Similarity



Meaning-level fit — survives vocabulary mismatch

Skill Match



Required skills covered by the candidate

Experience Match



Years of experience vs JD's stated range

Behavioral Signal



Leadership/ownership/impact language density

Platform Activity



GitHub / LinkedIn / StackOverflow signal

All weights are configurable in src/config.py — tune per role family.

Smart Use of LLMs — Only Where It Counts

✗ Naive Approach

- ✗ Send all 10,000 resumes through the LLM
- ✗ Cost: ~\$50-200 per run
- ✗ Latency: 30-60 minutes per batch
- ✗ LLM hallucination risk is load-bearing
- ✗ Impractical for real-world deployment

✓ Our Approach

- ✓ Embeddings + structured scoring ranks ALL candidates fast & cheap
- ✓ LLM sees ONLY the final top-25 shortlist
- ✓ Cost: <\$0.05 per run
- ✓ Latency: seconds
- ✓ LLM provides nuanced reasoning & re-rank — not blind trust

Result: LLM-quality judgment on the shortlist, embedding-scale efficiency on the full dataset.

Recruiters Can See Why — Not Just Who

output/ranked_candidates.csv

RANK	CANDIDATE	SCORE	SEMANTIC	SKILL MATCH	EXP MATCH	EXPLANATION
#1	Vikram Rao	91.2	0.89	100%	100%	Strong overall fit; covers all required skills; 9y fits well; shows ownership language
#2	Asha Mehta	82.7	0.81	93%	100%	Strong fit; covers most skills; 6y fits well; strong ownership/leadership language
#3	Karthik S.	67.4	0.73	80%	80%	Moderate fit; strong streaming background; 4y slightly under target

What We Built It With

SEMANTIC ENGINE

- sentence-transformers (all-mpnet-base-v2)
- scikit-learn (cosine similarity, TF-IDF fallback)

LLM LAYER

- Anthropic Claude (claude-sonnet-4-6)
- Used only for JD parsing + top-N re-rank

DATA PROCESSING

- pandas, numpy
- Schema-tolerant column mapping
- PyPDF2 / python-docx for file parsing

INTERFACE

- CLI via argparse (python -m src.main)
- Streamlit demo UI (app.py)

DESIGN PRINCIPLES

- No hard LLM API key dependency
- TF-IDF offline fallback if no internet
- All weights configurable in config.py

OUTPUT

- CSV with rank, scores, plain-English explanation
- LLM reasoning per candidate (optional)

The Outcome

A shortlist recruiters can actually trust.

- Understands the role — not just the keywords
- Sees the full candidate picture — skills, experience, behavior, platform
- Explains every ranking — auditable, not a black box
- Scales to 50k+ candidates cheaply — LLM only where it counts